

ground, one for the voltage and further wires if it is a stereo connection or has a mic. The important part of the working of a wired transmission medium is at its ends (the connectors), and those too will be covered in a subsequent chapter on connectors. For now, let us move on to the second category of transmission medium, the kind that really looks like magic – wireless transmission.

Wireless Audio

Whether you have an age-old MotoRokr bluetooth handset or a spanking new Apple Iphone 7, the sound your phone is playing reaches the device in your ears through the air, i.e Wirelessly. Again, note that though the medium is the same (air), this isn't the actual sound waves travelling but the encoded version of the audio information. To illustrate, let us travel even further back in time to visit one of the first major wireless mediums mankind has ever known – the Radio.

Surfing the radio waves

If you identify as a millennial, or even as a millennial's progeny, somewhere in your mind may lie a decaying memory that goes something like the following. You are speedily running your finger over the serrated edge of a submerged gear shaped dial, paying close attention to the shifting sound of white noise.

click

You spin the dial backwards, slowly. Like a whale from the depths of the ocean breaching the surface to spout, a voice or a song gradually emerges from the meaningless noise. Before you know it, the clarity of the station is submerging in the noise once more. Again, you reverse the direction of spin and find the sweet spot.

Of course, these days wherever there is radio (FM or AM) there is mostly also automatic station selection. Radio waves are electromagnetic waves of a comparatively high frequency (3kHz to 300GHz). They travel large distances via line of sight, as well as by reflection from the ionosphere.



If you owned one of these, perhaps you should start saving up for retirement.